LHAS: A Lightweight Network Based on Hierarchical Attention for Hyperspectral Image Segmentation

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***Abstract*— Deep learning has garnered extensive attention in hyperspectral image (HSI) processing. However, its application in HSI semantic segmentation tasks has been relatively limited. Although segmentation methods can often interpret images up to two orders of magnitude faster than classification methods when interpreting images of the same scene, the segmentation task requires the training data to be fully labeled, i.e., each pixel has a corresponding label. Such data are scarce in HSI data. To address this problem, this article proposes a lightweight segmentation network based on a hierarchical attention segmentation network (LHAS), in which a generalized data augmentation (GDA) method is utilized to acquire relatively sufficient data for semantic segmentation. Specifically, the hierarchical attention module is designed to extract global and local information on HSI patches from different layers. A prototype auxiliary module (PAM) of cluster contrast has also been developed to enhance feature discrimination. Across two different datasets in various scenarios, the proposed LHAS demonstrates superior segmentation performance compared to existing methods, affirming its effectiveness. In addition, experiments conducted on embedded devices validate the efficacy of LHAS.**

***Index Terms*— Generalized data augmentation (GDA), global and local feature extraction, hyperspectral image (HSI) semantic segmentation, lightweight network.**

I. Introduction

HYPERSPECTRAL image (HSI) consists of dozens or even hundreds of continuous bands, with each pixel capturing the reflectance spectral information of the corresponding ground object. The high spectral information in HSI allows for the distinguishing of various materials. Therefore, HSI has been applied in classification, segmentation, detection, and other fields [1], [2], [3], [4], [5]. However, due to the complexity of high-dimensional data and the limited label samples in HSI, there are significant challenges in the practical application of HSI-based algorithms.

In recent years, HSI classification methods based on neural networks have continuously emerged due to their powerful ability to extract discriminative features [6]. Nevertheless, several challenges hinder effective feature extraction and classification in HSI analysis. These challenges include the need to effectively fuse spectral and spatial features while managing the data complexity, enhancing the discrimination of feature representations, reducing computational costs, and mitigating sensitivity to spatial rotations. In addition, capturing homogeneous spectral features, exploiting bidirectional spectral correlations, and suppressing redundant spectral bands further complicate HSI processing, while preserving discriminative information with minimal computational overhead and understanding the mutual exclusion relationships between patterns remain critical issues.

To address the above challenges, a variety of approaches focusing on network architecture and feature extraction have been proposed. For instance, a weighted feature fusion approach combining convolutional neural networks (CNNs) and graph attention networks was introduced in [7] to improve the integration of weighted features. In efforts to enhance feature discrimination, Yu et al. [8] developed a spatial–spectral dense CNN framework with a feedback attention mechanism. Reducing the computational cost has been pursued via channel attention mechanisms that automatically design and optimize CNNs [9]. To counteract sensitivity to spatial rotations, a rotation-invariant attention network was proposed in [10]. In addition, Zhang et al. [11] addressed the challenge of capturing homogeneous features by developing a CNN-based spectral partitioning residual network. Moreover, to fully exploit spectral correlation, Mei et al. [12] proposed a bidirectional long short-term memory (Bi-LSTM) network to leverage both forward and backward dependencies. The issue of redundant spectral bands was tackled by developing a cross-attention spectral–spatial network in [13], while Liu et al. [14] enhanced the representation of target pixels with a center attention network. Preserving discriminative information while minimizing computational complexity was achieved through a mask-reconstruction-based decoupled convolution network (MrDCN) [15]. Furthermore, Gao et al. [16] introduced a learnable pattern sampling module to effectively

Received 16 July 2024; revised 23 September 2024 and 12 February 2025; accepted 24 February 2025. Date of publication 12 March 2025; date of current version 24 March 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62301044, in part by Beijing Natural Science Foundation under Grant 4254072, and in part by the University-Industry Collaborative Education Program under Grant 230800001302636. *(Corresponding author: Yunhao Gao.)*

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Digital Object Identifier 10.1109/TGRS.2025.3547824

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capture mutual exclusion relationships, and a cross-scale mixing attention transformer was proposed to capture long-range dependencies [17].

Meanwhile, other approaches have been proposed to tackle data-related challenges and enhance practical applicability. Miao et al. [18] proposed a novel method to solve the class-imbalanced issue by performing remarkable efforts in pseudo-label selection from class-imbalanced data. To alleviate the need for large annotated datasets and reduce network complexity in HSI classification, SAFDNet was proposed in [19], which leverages stochastic adaptive Fourier decomposition for unsupervised feature extraction. To address haze degradation in HSIs, Fu et al. [20] developed a novel synthesis strategy to construct the HyperDehazing benchmark dataset and proposed HyperDehazeNet, which effectively removes complex haze with high spectral reconstruction quality and low computational cost. In addition, to enhance model generalization, Zhang et al. [21] designed a language-aware domain generalization network to learn cross-domain-invariant representations, and a single-source domain expansion network was proposed in [22] to learn domain-invariant information.

However, the above classification performance is still limited by the real-time requirements of tasks, especially with limited computing power on edge devices. In contrast, the interpretation process of HSI segmentation requires only one inference or patch-by-patch processing, which gives semantic segmentation significant advantages in inference speed. In the field of HSI segmentation, existing semantic segmentation methods face several challenges. The high dimensionality of spectral data and the complex spatial structures in HSI make it difficult for current models to effectively fuse spectral and spatial information. For instance, the segmenter model developed by Strudel et al. [23] used a transformer to model global context, and the STDC network proposed by Fan et al. [24] aggregated multiscale information, but both struggle with the intricate characteristics of HSI. Moreover, the imbalance and scarcity of labeled samples in HSI data further complicate the segmentation process. To address the limited labeled data, Zhou et al. [25] introduced a nonparametric method based on nonlearnable strategies, and similarly, Li et al. [26] proposed a hierarchical semantic segmentation (HSS) model to generate well-structured pixel representations that enhance the capture of complex features in HSI.

Recent advances in prototype and contrastive learning have also been explored to improve HSI segmentation performance. Wang et al. [27] proposed a pixel-wise contrastive algorithm that enhances the similarity of same class points, and Li et al. [28] developed a language-driven semantic image segmentation method to generalize to unseen categories, further improving segmentation outcomes. Considering the need for computational efficiency and real-time processing, Guo et al. [29] employed a simple CNN architecture named SegNext combined with efficient convolution operations to reduce computational costs, while Peng et al. [30] proposed a lightweight model for real-time semantic segmentation with simple pyramid pooling module (PP-LiteSeg) to aggregate information with minimal computational overhead.

Although semantic segmentation methods can achieve real-time image category inference, there is a significant lack of work on HSI semantic segmentation. Several issues persist.

1) HSI semantic segmentation requires annotating the training data for the entire image, and fine annotation of the entire image is a difficult problem with high costs to solve.
2) For HSI semantic segmentation, the distribution of the number of different ground objects is usually unbalanced. The categories with more samples collected can provide more information for the network, while the categories with fewer samples may not provide enough information for the network.

To address these challenges, this article presents generalized data augmentation (GDA). By establishing a database from existing annotated data, we generate an expanded and sufficient training set using four methods. We introduce a lightweight segmentation network based on a hierarchical attention segmentation network (LHAS), designed to extract features from images both globally and locally. Unlike the conventional encoder–decoder architecture, LHAS employs only an encoder structure. It utilizes scale attention to weight and directly feeds semantic features from different layer scales into the segmentation head, thereby reducing the computational load in the decoding phase. These enhancements aim to improve segmentation accuracy and reduce computational complexity, making the model more suitable for real-world deployment. In addition, we propose an auxiliary classifier, prototype auxiliary module (PAM), based on prototype learning, aimed at enhancing feature separability in the semantic space. Extensive experimental results have shown that the proposed LHAS is ahead of related segmentation frameworks.

The main contributions of this work are summarized as follows.

1) A GDA technique is proposed, which provides sufficient training data for network training.
2) A plug-and-play auxiliary clustering module was designed to enhance feature separability in the semantic space.

The rest of this article is organized as follows. Section II explicitly explains the proposed data augmentation and LHAS. Section III displays the experimental results to demonstrate the superior performance of LHAS. In Section IV, we conclude this article.

II. Proposed Lightweight Network Based on Hierarchical Attention

The proposed LHAS is illustrated in Fig. 2, which includes a simple feature extraction stem and a prototype learning clustering module for assisted segmentation.

A. Generalized Data Augmentation

GDA can be categorized into two components. Initially, all labeled pixels within the original 3-D tensor are categorized based on their labels. Each category of pixels is then aggregated into a separate dictionary, which is utilized for

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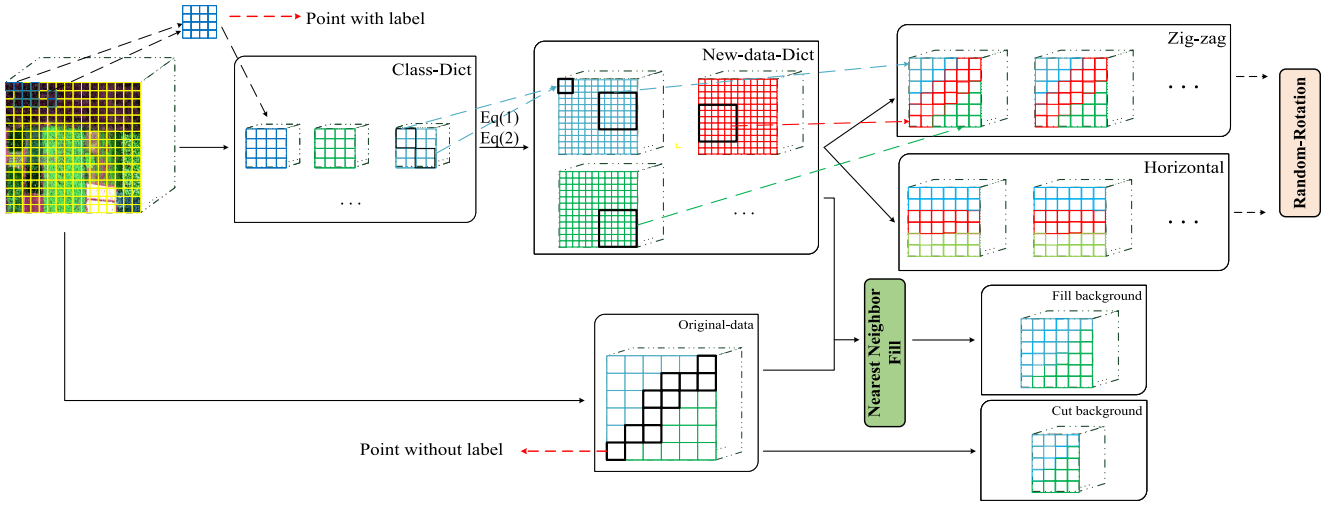


Fig. 1. Illustration of generating new data and four illustrations of adding new data. The left part illustrates the process of generating new data, while the right part presents four different methods for adding new data. While maintaining the spatial relative positions of labeled data in the original dataset, the dictionary for each category is obtained by aggregation. Using the obtained dictionary, (1) and (2) are used to calculate the similarity, and new data are generated to establish a dictionary of new data. Three categories are randomly selected from the new data dictionary, and the blocks are spliced in zigzag and horizontal directions to construct the trainable HSI. Meanwhile, unlabeled points in the original image can be handled in two ways. One approach is to assign the nearest spatial category to the unlabeled points using the new data dictionary, while the other is to simply remove these points, both leading to a trainable HSI.

generating new data. Subsequently, four data augmentation methods are applied to the existing data, expanding the dataset to alleviate the inadequacy of labeled HSI semantic segmentation data. Once the expanded dataset is obtained, it is partitioned into training and testing sets to facilitate the optimization network.

1) New Data Generation: As shown in Fig. 1, during the new data generation phase, the point with the same category label is extracted from the original image to form a new 3-D tensor. The new 3-D tensors are then constructed based on these existing categories. It is worth noting that the pixels' relative positions in the obtained new 3-D tensor are consistent with the original image. Each 3-D tensor serves as a dictionary for the respective class (Class-Dict) and is used for generating new pixels. The generation of new pixels is defined as follows:

$$\alpha_{ij} = e^{-\frac{\|x_i - x_j\|^2}{\sigma^2}} + \frac{x_i \cdot x_j}{\|x_i\| \cdot \|x_j\|} \tag{1}$$

$$\bar{X}_i = (1 - \beta \cdot \alpha_{ij}) \cdot X_i + \beta \cdot \alpha_{ij} \cdot X_j \tag{2}$$

where α_{ij} represents the affinity factor, obtained as the average of two similarity calculations. The parameters σ and β are hyperparameters. x_i and x_j are the mean of two randomly selected pixel block X_i and X_j from the same dictionary. $\bar{X}_i$ denotes the newly generated pixel block. Since the two-pixel blocks selected for generating a new pixel block originate from the same class dictionary and are employed in computing the affinity factor, the newly generated pixel is regarded as a linear combination of pixels associated with the same class. Consequently, these pixels inherently pertain to this selected class and can be categorized accordingly. Those generated blocks of pixels form a larger category dictionary called New-data-Dict, which will be used for subsequent data augmentation.

2) Four Kinds of Data Augmentation: As depicted in Fig. 1, GDA comprises four methods designed to augment HSI.

As shown in the top-right part of Fig. 1, following the pattern of feature distribution, where different points edges intersect, the first and second data augmentation methods involve stitching pixels in a zigzag and horizontal pattern, respectively. After the stitching process is completed, a new 3-D pixel block is obtained. This block effectively simulates the boundary scenarios of most ground objects through random rotation, flipping, and other operations.

Below the previous two methods in Fig. 1, the third data augmentation method considers the distribution pattern of ground objects, specifically the tendency of similar ground objects to cluster. Pixels without labels in the original 3-D image block are selected, and the nearest labeled pixel serves as a reference. The corresponding dictionary is chosen based on the label, and the unlabeled pixel is replaced by a random pixel from the dictionary of the corresponding category. This ensures that each point in the 3-D block has a label, making it suitable for training.

As shown in the bottom-right part of Fig. 1. The fourth data augmentation strategy involves simply removing pixels without labels. The remaining pixels maintain their original relative position and are consolidated into smaller 3-D tensors, thereby preserving the topological information of the original data to the greatest extent possible.

Four data augmentation methods are used to generate new data and augment the original labeled data in four ways: horizontal and zigzag direction stitching pixels, filling, and removing the background. Together with random rotation, flipping, and other subsequent operations, they are sufficient to cover most of the distribution and form a dataset enough for training.

B. Hierarchical Attention Based Segmentation Network

As depicted in Fig. 2, the proposed LHAS comprises a stem module, two layers of para-block, and a plug-and-play

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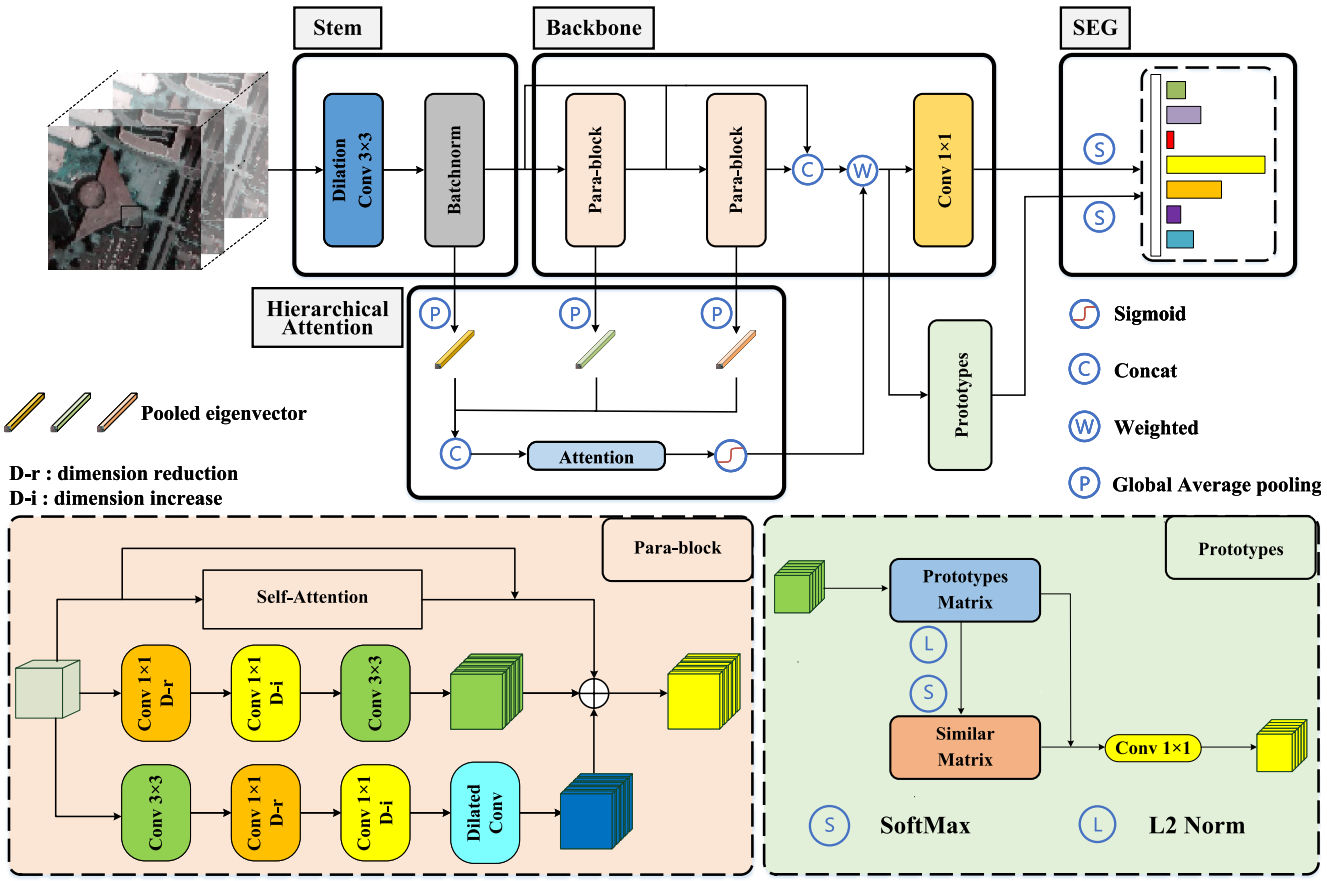


Fig. 2. Illustration for the proposed LHAS, which includes a stem and backbone, hierarchical attention module, and PAM.

auxiliary clustering module. Following feature extraction, features of different levels are fused through hierarchical attention and directly fed into the segmentation head via a 1×1 convolution. This design choice contributes to computational and operational time savings.

1) Hierarchical Attention: As illustrated in Fig. 2, the backbone is mainly composed of two para-blocks. The para-blocks are employed to extract global and local dependencies simultaneously. Here, the global dependencies are implemented through attention, and local dependencies are supplemented through a two-branch convolution combination. To integrate feature information from different levels, cross-layer information interaction is facilitated using hierarchical attention.

Specifically, the features from the stem module and two para-blocks F_1–F_3 are transformed to 1-D vectors by the global average pooling (GAP) and then concatenate to obtain the integrated features as follows:

$$F = \text{Concat}(\text{GAP}(F_1, F_2, F_3)) \tag{3}$$

where Concat$(\cdot)$ denotes the concatenation operation along the channel dimension. After that, a self-attention module is utilized to explore the correlation of integrated features. The correlation is calculated as follows:

$$F_c = \text{SoftMax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{4}$$

where $Q = F * W_1, K = F * W_2, V = F * W_3$, and W_1–W_3 are the weight matrix of 1-D convolution. Therefore, the hierarchical features are obtained as follows:

$$F_h = \text{Concat}(F_1, F_2, F_3) \odot F_c. \tag{5}$$

$\odot$ is multiplication by channel. Finally, the 1×1 convolution is employed to convert dimensions into the number of categories and then fed them into the semantic segmentation head to generate the segmentation map. The loss function is

$$\text{Loss} = \sum_{i=1}^{I}\sum_{n=1}^{N} y_{i,n}\log(p_{i,n}) \tag{6}$$

where I is the number of image samples; N is the number of classification categories; and y and p are the label and prediction probability, respectively.

C. Prototype Auxiliary Module

Considering the fewer sample points of some categories in HSI data, this limitation affects the performance of semantic segmentation. Categories with fewer samples provide insufficient information during the training process and are easily misclassified by the network into other categories. Inspired by recent research results in prototype learning and contrastive learning methods, clustering and contrastive learning through prototype learning can enhance the separability of each class and increase the diversity of different classes, thus further improving the network performance. The PAM cluster contrast module is designed for HSI semantic segmentation tasks.

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As shown in Fig. 2, the PAM module is used to precluster the extracted features, and then, the clustered features are input to the segmentation head. The prototype matrix is $P \in R^{l\times k}$, in which k is the cluster center and l is the number of channels. Then, the similarity between features and cluster centers is calculated by dot multiplication. The similarity is characterized by the $L2$ norm and SoftMax with a temperature coefficient. The clustering features can be obtained by multiplying the obtained features with the clustering similarity matrix

$$S = \text{SoftMax}(L2(XP)) \tag{7}$$

where S is the similar matrix, X is the input feature, $L2$ is the $L2$ norm, and SoftMax is the nonlinear active function. The clustering feature is obtained by multiplication of similarity matrix and the prototype matrix: CF = PS, where CF is the clustering feature. The loss function of PAM is

$$\text{Loss} = \sum_{i=1}^{I}\sum_{n=1}^{N} y_{i,n}\log(p_{i,n}) \tag{8}$$

where I is the number of image samples; N is the number of classification categories; and y and p are the label and prediction probability, respectively.

Learnable prototype matrix P online updates features and cluster centers, thus avoiding two-stage training. It is worth noting that the selection of cluster centers is usually larger than the number of categories because the data labels may not be finely defined, leading to potential subdivision and spectral curve differences within the same category.

The entire PAM module is dedicated during training to support clustering for semantic segmentation. In the actual scene inference process, it can be entirely omitted to improve inference speed.

D. Analysis on the Proposed Method

For HSI pixel-level semantic segmentation, spatial and spectral features convey distinct information. Spectral information reflects the discriminative characteristics of substances, while spatial information depicts the geometric attributes of different materials and their spatial distribution. Consequently, the fusion of rich features between space and spectrum is highly advantageous for leveraging semantic segmentation methods in interpretation. However, deep learning-based semantic segmentation methods often face the challenge of insufficient labeled data for HSI, necessitating data expansion and label supplementation through GDA.

For semantic segmentation tasks, the commonly used encoder–decoder structure processes information at different scales and depths in the encoder and decoder, respectively, and completes feature extraction. However, encoders and decoders often require comparable computational resources, posing challenges for deploying algorithms in production environments. Therefore, LHAS adopts a network design with 1×1 convolution as the decoder for end-to-end processing and uses hierarchical attention to realize the interaction of features at different depths. Meanwhile, attention, small convolution, and dilated convolution are used in the para-block to extract local and global information, which facilitates the extraction of discriminative details from HSI data.

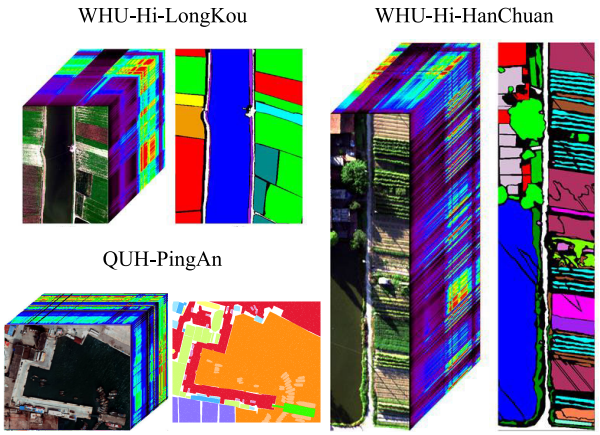


Fig. 3. Maps of labeled samples distributions of WHU-Hi-LongKou, WHU-Hi-HanChuang, and QUH-PingAn datasets.

The data available for semantic segmentation in HSI are much less than the data available for semantic segmentation in visible light and multispectral images. There is a gap of orders of magnitude in sample size between some categories with smaller and more samples. This limitation prevents the full exploitation of the advantages of HSIs. Recent work on prototype and contrastive learning has shown that this problem can be partially mitigated by clustering [31]. Therefore, LHAS introduces the PAM module to cluster and learn the proposed features to enhance the similarity between similar features and highlight the differences between different categories of features. This enhancement enables better extraction of discriminative information from the HSI.

In addition, including the PAM module increases training time and computational complexity, contributing to a substantial proportion of network parameters. However, in real-world deployment scenarios, the network can be simplified by omitting PAM and utilizing only the feature extraction module.

III. Experiments and Analysis

In this section, the effectiveness of the proposed LHAS is evaluated using three HSI datasets. First, the description of considered HSI datasets is introduced, which are collected from three cities (WHU-Hi-LongKou [32], WHU-Hi-HanChuan, and QUH-PingAn [33]). Second, the ablation experiments are conducted to validate the effectiveness of the proposed LHAS. Subsequently, comparative experiments are conducted on other related works to verify the superior performance of the proposed model. Finally, we conducted simulation experiments on embedded devices to verify the adaptability.

A. Datasets

1) WHU-Hi-LongKou Dataset: The WHU-Hi-LongKou dataset was captured on July 17, 2018, in Longkou, Hubei, China. It was collected using a DJI Matrice 600 Pro (DJI M600 Pro) UAV platform equipped with an 8-mm focal length Headwall Nano-Hyperspec imaging sensor. The weather conditions during data acquisition were clear and without clouds, with a temperature of approximately 36 °C and a relative air humidity

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TABLE I
CLASS LABELED SAMPLE DISTRIBUTION OF WHU-HI-LONGKOU

Class No.	Class Name	points
1	Corn	34511
2	Cotton	8374
3	Sesame	3031
4	Broad-leaf soybean	63212
5	Narrow-leaf soybean	4151
6	Rice	11854
7	Water	67056
8	Roads and houses	7124
9	Mixed weed	5229
Total		204542

TABLE II
CLASS LABELED SAMPLE DISTRIBUTION OF WHU-HI-HANCHUAN

Class No.	Class Name	points
1	Strawberry	44735
2	Cowpea	22753
3	Soybean	10287
4	Sorghum	5353
5	Water spinach	1200
6	Watermelon	4533
7	Greens	5903
8	Trees	17978
9	Grass	9469
10	Red roof	10516
11	Gray roof	16911
12	Plastic	3679
13	Beare soil	9116
14	Road	18560
15	Bright object	1136
16	Water	75401
Total		257530

of about 65%. The study area consisted of a straightforward agricultural scene featuring six different crop species: corn, cotton, sesame, broad-leaf soybean, narrow-leaf soybean, and rice. The UAV flew at an altitude of 500 m, capturing imagery sized at 550 × 400 pixels. This dataset comprises 270 bands ranging from 400 to 1000 nm, with the spatial resolution of the UAV-borne hyperspectral imagery being approximately 0.463 m. Fig. 3 and Table I offer an overview of this dataset.

2) WHU-Hi-HanChuan Dataset: The WHU-Hi-HanChuan dataset, obtained on June 17, 2016, between 17:57 and 18:46 in Hanchuan, Hubei, China, was captured using a Leica Aibot X6 UAV V1 platform equipped with a 17-mm focal length Headwall Nano-Hyperspec imaging sensor. The data collection occurred under clear and cloudless weather conditions with temperatures around 30 °C and approximately 70% relative air humidity. This dataset covers a rural–urban fringe area containing various elements such as buildings, water bodies, and cultivated land hosting seven crop species, namely, strawberry, cowpea, soybean, sorghum, water spinach, watermelon, and greens. Captured at an altitude of 250 m, the imagery dimensions are 1217 × 303 pixels, comprising 274 bands ranging from 400 to 1000 nm. The spatial resolution of the hyperspectral imagery from the UAV is approximately 0.109 m. It is important to note that this dataset was collected during the afternoon when the solar elevation angle was low, resulting in numerous shadow-covered areas across the image. Refer to Fig. 3 and Table II for an overview of this dataset.

TABLE III
CLASS LABELED SAMPLE DISTRIBUTION OF QUH-PINGAN

Class No.	Class Name	points
1	Ship	48935
2	Seawater	578113
3	Trees	8345
4	Concrete stucture	88973
5	Floating pier	20759
6	Brick houses	14086
7	Steel houses	13991
8	Wharf construction land	83113
9	Car	8108
9	Road	5229
Total		1140937

TABLE IV
DETAILS OF DATASET PARTITIONING AFTER GDA

dataset	Training	Testing	Validating
WHU-Hi-Longkou	1892	10738	63
WHU-Hi-HanChuan	1886	11170	100
QUH-PingAn	1900	10960	320

3) QUH-PingAn Dataset: The QUH-PingAn dataset was obtained on May 19, 2021, during the period from 10:43 to 11:28 at the Huangdao Pingan Passenger Ship Terminal in Qingdao, China. During data acquisition, a UAV was employed at a flight altitude of 200 m to capture hyperspectral imagery with an approximate spatial resolution of 0.10 m. The collected imagery covers an area of 1230 × 1000 pixels and comprises 176 spectral bands spanning wavelengths from 400 to 1000 nm. This dataset distinguishes itself by classifying the scene into ten land cover categories: ship, seawater, trees, concrete structure building, floating pier, brick houses, steel houses, wharf construction land, car, and road. Despite an overall balanced distribution of these classes, the targets exhibit notable variations in scale. Specifically, seawater and road represent large-scale features; trees, floating pier, brick houses, and steel houses fall into a medium scale, whereas ship and car are examples of small-scale objects. For further details, please refer to Fig. 3 and Table III. The training, test, and validation sets of the three datasets are shown in Table IV. The Training and Testing are the data generated by data augmentation for network training and parameter tuning. The Validating is the original data, which is used to verify the algorithm performance.

B. Parameter Analysis

The core parameter for LHAS is β, which determines the construction of new data in GDA. Because the effect of β is symmetric around 0.5, β is selected from 0.1, 0.2, 0.3, 0.4, 0.5. As shown in Table VI, with the change of β, the similarity of the new data to the two existing data changes. The new data generated will be different from the existing data, bringing more features for gradient propagation, and the segmentation accuracy will be higher. The accuracy reaches its highest when β is 0.2. When the mask ratio continues to increase, the new

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TABLE V

ANALYSIS EXPERIMENTS ABOUT THE SUGGESTED PB (PARA-BLOCK), HAT, PAM, GDA, AND HA IN REFERENCE TO THE SEGMENTATION PERFORMANCE [OA (%) AND MIOU (%)] ON HANCHUAN AND LONGKOU DATASETS

PB	HAT	PAM	GDA	HA	OA			MIOU		
					HanChuan	LongKou	PingAn	HanChuan	LongKou	PingAn
✗	✗	✗	✗	✗	67.19	71.67	70.88	48.21	54.44	51.99
✗	✗	✗	✗	✓	71.44	76.15	75.53	50.40	58.96	57.49
✗	✗	✗	✓	✓	81.88	93.06	88.89	62.99	75.13	68.02
✓	✗	✗	✓	✓	82.52	95.12	90.84	64.50	84.08	72.20
✗	✓	✗	✓	✓	83.28	95.97	91.15	66.10	84.22	72.24
✗	✗	✓	✓	✓	84.63	96.84	92.02	67.35	86.73	76.29
✓	✓	✗	✓	✓	85.67	97.95	94.78	69.06	92.83	80.33
✓	✗	✓	✓	✓	84.31	97.81	94.31	68.13	90.18	79.06
✗	✓	✓	✓	✓	86.94	98.97	95.16	70.41	94.96	81.74
✓	✓	✓	✓	✓	**86.95**	**99.10**	**95.75**	**70.49**	**96.60**	**83.07**

TABLE VI

RELATIONSHIP BETWEEN β, OA, AND MIOU (%) ON THE HANCHUAN AND LONGKOU DATASETS

	Dataset	β				
		0.1	0.2	0.3	0.4	0.5
OA	HanChuan	86.17	**86.95**	84.59	85.17	84.33
	LongKou	98.32	**99.10**	97.53	95.46	95.37
	PingAn	94.88	**95.75**	93.90	92.21	92.77
MIOU	HanChuan	69.81	**70.49**	66.99	68.93	66.12
	LongKou	92.75	**96.60**	90.03	84.26	84.11
	PingAn	80.61	**83.07**	79.57	76.33	77.07

TABLE VII

ANALYSIS OF THE EFFECTIVENESS OF PAM WITH DIFFERENT METHODS IN THE HANCHUAN DATASET

Method	OA		MIOU	
	✗	✓	✗	✓
Deep-v3	77.35	78.11	54.63	55.72
MASK2Former	73.08	74.16	51.02	52.37
PID	80.81	82.24	60.55	63.98
PSP	73.12	76.91	51.02	53.66
TopFormer	85.13	85.85	68.24	68.51
SEGnet	86.31	86.47	69.68	70.02

TABLE VIII

ANALYSIS OF THE EFFECTIVENESS OF PAM WITH DIFFERENT METHODS IN THE LONGKOU DATASET

Method	OA		MIOU	
	✗	✓	✗	✓
Deep-v3	94.26	95.14	77.50	80.13
MASK2Former	92.19	93.89	70.66	73.11
PID	94.12	94.91	77.01	77.70
PSP	93.01	93.83	74.78	75.06
TopFormer	97.06	98.33	84.17	90.06
SEGnet	96.90	97.86	87.00	90.01

data are close to a simple overlay of the existing data. The accuracy gain is reduced.

C. Ablation Studies

The ablation experiment is conducted to validate the effectiveness of the proposed GDA, para-block, hierarchical attention, and PAM. Classification performance is measured using overall accuracy (OA) and mean intersection over union (MIOU). As listed in Table V, after GDA, para-block, hierarchical attention, and PAM are added into HSI segmentation, a great improvement is achieved in segmentation performance. In LHAS, the OA and MIOU increased by 15.51%, 22.95%, and 20.22%, and 20.09%, 37.36%, and 25.58% for HanChuan, LongKou, and PingAn datasets, respectively. Hence, the effectiveness of GDA, para-block, hierarchical attention, and PAM for HSI segmentation is confirmed.

As reported in Table V, after the GDA operation, there have been obvious advances in segmentation performance. In LHAS, the OA and MIOU increased by 10.44%, 16.91%, and 13.35%, and 12.59%, 16.17%, and 10.53% for HanChuan, LongKou, and PingAn datasets, respectively. Hence, the effectiveness of GDA for HSI segmentation is confirmed.

To discuss the effect of the para-block, the traditional convolution is used to replace the para-block of LHAS for analysis experiments, and the effect of the para-block is further discussed. In LHAS, the OA and IOU increased by 0.64%, 2.06%, and 1.95%, and 1.51%, 8.95%, and 4.18% for HanChuan, LongKou, and PingAn datasets, respectively. Hence, the effectiveness of the para-block is confirmed again.

As listed in Table V, after hierarchical attention and PAM are added into LHAS, there have been advances in segmentation performance. In LHAS, the OA increased by 1.40%, 2.91%, and 2.26%, and 2.75%, 3.78%, and 3.13%, and the MIOU increased by 3.11%, 9.06%, and 4.22%, and 4.36%, 11.60%, and 8.27% for HanChuan, LongKou, and PingAn datasets, respectively. Fig. 4(a) shows the visual feature map of HanChuan without PAM and Fig. 4(d) shows the visual feature map of HanChuan with PAM. Fig. 4(b) shows the visual

TABLE IX

ANALYSIS OF THE EFFECTIVENESS OF PAM WITH DIFFERENT METHODS IN THE PINGAN DATASET

Method	OA		MIOU	
	✗	✓	✗	✓
Deep-v3	93.69	94.02	75.34	77.67
MASK2Former	91.62	92.73	70.61	72.81
PID	92.47	94.27	73.12	75.16
PSP	88.80	91.12	63.17	68.02
TopFormer	94.38	94.93	78.19	80.14
SEGnet	94.28	95.38	78.42	81.98

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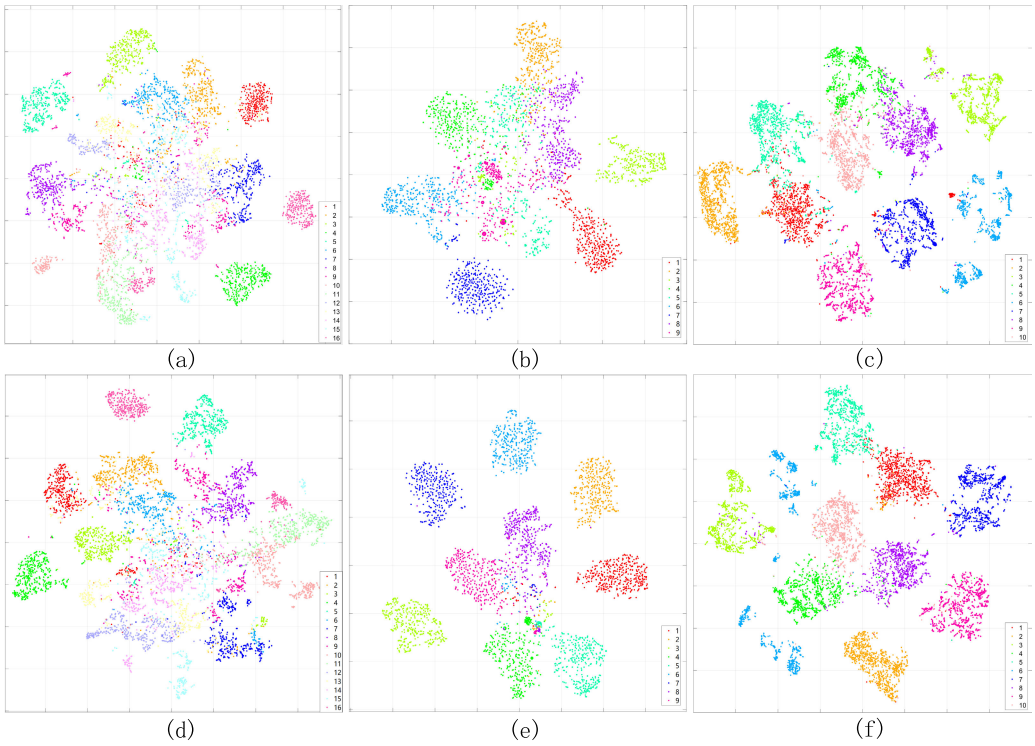


Fig. 4. Visual feature map of LHAS without and with PAM. (a) HanChuan without PAM. (b) LongKou without PAM. (c) PingAn without PAM. (d) HanChuan with PAM. (e) LongKou with PAM. (f) PingAn with PAM.

TABLE X
CLASSIFICATION AND MIOU RESULTS (%) ON THE HANCHUAN DATASET WITH DIFFERENT METHODS

Class No.	Deep-v3 [34]	MASK2Former [35]	PID [36]	PSP [37]	TopFormer [38]	SEGnet [39]	LHAS
1	78.14	83.88	86.14	84.24	84.60	84.34	87.71
2	64.47	74.28	74.76	73.78	79.72	81.00	83.91
3	85.07	91.14	92.38	91.38	95.08	96.01	96.39
4	96.92	97.29	95.39	97.44	98.94	98.82	98.97
5	94.25	83.33	87.25	86.42	96.67	95.17	96.75
6	65.65	71.19	77.45	71.17	72.38	70.77	69.69
7	43.52	39.01	36.66	37.47	61.44	64.75	70.90
8	88.99	70.05	86.23	70.39	90.49	91.34	91.23
9	47.06	38.91	51.95	38.43	59.31	58.25	60.74
10	39.33	36.34	56.98	36.27	63.47	72.05	76.11
11	30.94	90.25	34.73	89.99	65.63	75.24	74.39
12	68.44	32.64	30.52	34.57	60.04	64.37	65.56
13	85.95	76.25	86.09	78.95	82.17	85.48	84.48
14	75.19	57.07	77.85	56.35	79.87	80.50	79.08
15	80.11	71.30	70.42	72.54	74.56	73.06	74.74
16	98.05	76.58	99.10	76.46	99.57	98.95	97.47
OA	77.35	73.08	80.81	73.12	85.13	86.31	86.95
AA	71.38	68.10	71.49	68.49	79.00	80.63	81.76
Kappa*100	73.54	69.09	77.28	69.15	82.40	83.87	84.66
MIOU	54.63	51.02	60.55	51.02	68.24	69.68	70.49

feature map of LongKou without PAM and Fig. 4(e) shows the visual feature map of LongKou with PAM. Fig. 4(c) shows the visual feature map of PingAn without PAM and Fig. 4(f) shows the visual feature map of PingAn with PAM. It can be found that the PAM can increase the feature separability. Hence, the effectiveness of hierarchical attention and PAM for HSI classification is confirmed.

As reported in Table V, to further illustrate the applicability of HA, the attention operations in HA are replaced by ordinary 1-D convolution and pooling operations without interaction of multiscale information. After the HA operation, there have been advances in segmentation performance. In LHAS, the OA and MIOU increased by 4.25%, 4.48%, and 4.65%, and 2.19%, 4.52%, and 5.5% for HanChuan and LongKou datasets.

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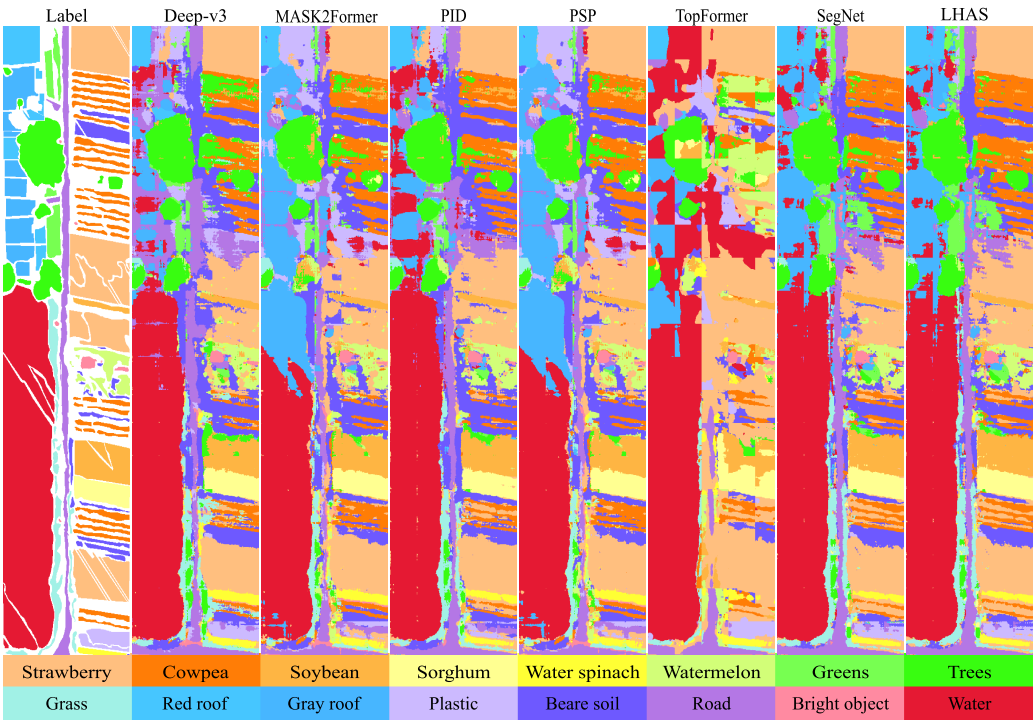


Fig. 5. HanChuan's ground-truth and segmentation maps in different methods.

TABLE XI
CLASSIFICATION AND MIOU RESULTS (%) ON THE LONGKOU DATASET WITH DIFFERENT METHODS

Class No.	Deep-v3	MASK2Former	PID	PSP	TopFormer	SEGnet	LHAS
1	99.25	96.42	99.14	95.61	98.19	96.94	99.71
2	82.31	87.38	81.92	88.40	96.35	91.35	97.90
3	80.90	83.07	80.11	85.48	78.32	92.74	99.27
4	92.84	91.89	92.79	87.95	98.21	95.76	98.94
5	93.52	79.84	93.74	95.09	74.66	94.41	97.28
6	78.53	84.51	76.67	85.84	97.31	95.08	99.92
7	99.98	99.98	99.97	99.98	99.99	99.98	99.99
8	92.03	57.10	91.89	85.26	94.68	91.93	97.74
9	71.45	56.34	72.54	84.39	70.53	94.95	97.16
OA	94.26	92.19	94.12	93.01	97.06	96.90	99.31
AA	87.87	81.83	87.64	89.78	89.80	94.79	98.66
Kappa*100	92.50	89.80	92.31	90.93	96.13	95.94	99.10
MIOU	77.50	70.66	77.01	74.78	84.17	87.00	96.60

Hence, the effectiveness of HA for HSI segmentation is confirmed.

In particular, GDA brings the largest accuracy boost. This is likely because generating new data feeds different gradients back and forth during network training, helping the network to better tune its parameters. This may indicate that the segmentation task of the network is not a lack of information, but a lack of data diversity for training with different return gradients. When training networks for segmentation tasks, more consideration should be given to more efficient training with insufficient data. Therefore, as a simple data augmentation method, GDA can help the network achieve better segmentation results.

D. PAM Performance Analysis

PAM is a plug-and-play module. This experiment is conducted to validate the effectiveness of the PAM on other segment methods. As listed in Table VII, after PAM is added into segment methods, there have been advances in segmentation performance. In the HanChuan dataset, the OA increased by 0.76%, 1.08%, 0.43%, 3.79%, 0.72%, and 0.16%, and the MIOU increased by 1.09%, 1.35%, 3.43%, 2.62%, 0.27%, and 0.34% for Deep-v3, MASK2Former, PID, PSP, TopFormer, and SEGnet, respectively. As listed in Table VIII, after PAM is added into segment methods, there have been advances in segmentation performance. In the LongKou dataset, the OA increased by 0.88%, 1.70%, 0.79%, 0.82%, 1.27%, and 0.96% and the MIOU increased by 2.63%, 2.45%, 0.69%, 0.28%, 5.89%, and 3.01% for Deep-v3, MASK2Former, PID, PSP, TopFormer, and SEGnet, respectively. As listed in Table IX, after PAM is added into segment methods, there have been advances in segmentation performance. In the PingAn dataset, the OA increased by 0.33%, 1.11%, 1.8%, 2.32%, 0.55%, and 1.1% and the MIOU increased by 2.33%, 2.2%, 2.04%, 4.85%,

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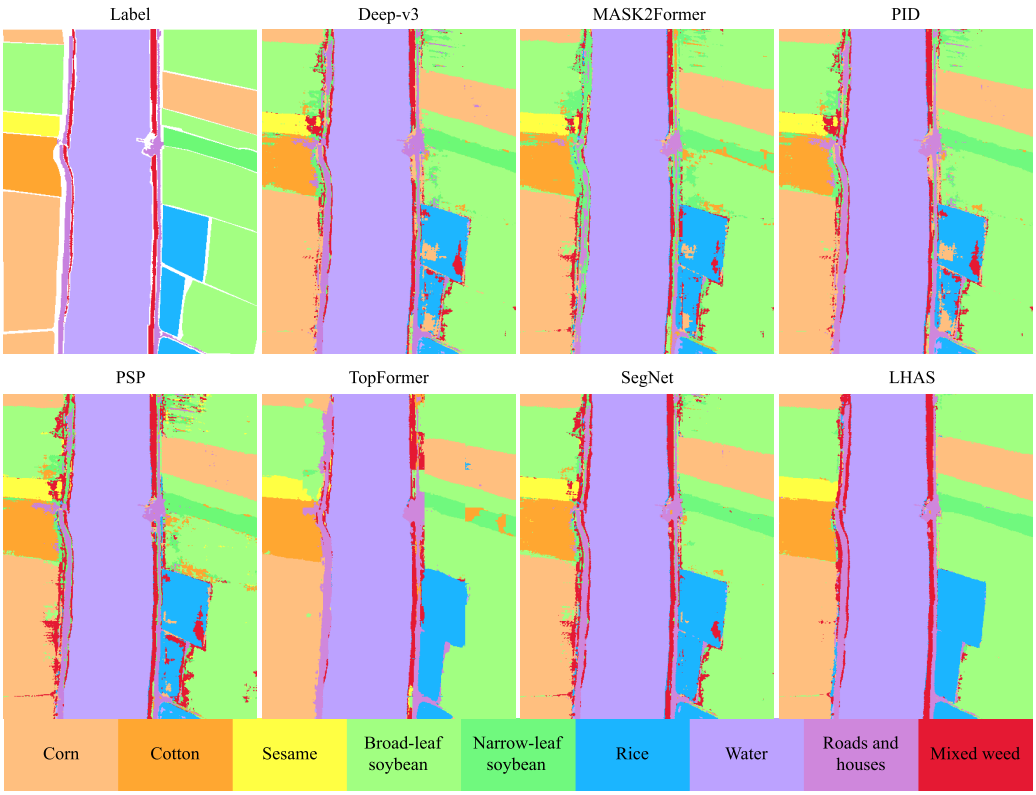


Fig. 6. LongKou's ground-truth and segmentation maps in different methods.

TABLE XII
CLASSIFICATION AND MIOU RESULTS (%) ON THE PINGAN DATASET WITH DIFFERENT METHODS

Class No.	Deep-v3	MASK2Former	PID	PSP	TopFormer	SEGnet	LHAS
1	87.23	82.84	87.04	71.72	92.54	90.40	94.33
2	96.77	96.97	97.04	96.07	97.00	97.25	97.06
3	99.67	95.79	97.89	95.18	99.93	99.85	99.98
4	85.27	76.53	84.59	62.25	92.28	88.09	95.53
5	90.62	86.36	87.16	74.13	95.09	95.26	98.45
6	97.71	92.64	97.57	92.13	99.14	98.29	99.02
7	97.00	96.55	97.73	91.86	98.14	97.62	99.49
8	89.87	84.90	89.21	74.51	91.21	89.55	93.35
9	91.84	94.68	95.69	81.61	95.68	96.70	99.60
10	91.97	88.74	86.99	90.24	90.17	91.49	93.27
OA	93.69	91.62	92.47	88.80	94.38	94.28	95.75
AA	92.80	89.60	92.09	82.97	95.12	94.45	97.01
Kappa*100	90.73	87.70	88.96	83.50	91.75	91.59	93.76
MIOU	75.34	70.61	73.12	63.17	78.19	78.42	83.07

1.95%, and 3.56% for Deep-v3, MASK2Former, PID, PSP, TopFormer, and SEGnet, respectively.

E. Segmentation Performance

In Tables X–XII, the class-specific accuracy (CA), OA, average accuracy (AA), Kappa coefficient, MIOU, and intersection over union (IOU) are used to evaluate the segmentation performance. Since there are few works on the semantic segmentation of HSIs, the comparison methods are selected from the latest works that perform well in visible light semantic segmentation. These include DeepLabv3, Mask2Former, PID, PSPNet, TopFormer, and SegNet, where TopFormer and SegNet consistently achieve good semantic segmentation scores on three datasets. The suggested LHAS performs better and more effectively than all state-of-the-art techniques listed above by approximately 0.64% and 2.97% in HanChuan, 2.25% and 9.60% in LongKou, and 1.37% and 4.65% in PingAn, surpassing the second-highest methods in OA and MIOU.

To visually display LHAS's effectiveness, segmentation maps of all the methods listed above are displayed in Figs. 5–7. The white on the label indicates the unlabeled area. The results of DeepLabv3 contain a lot of salt and pepper noise, and the worst segmentation results are shown

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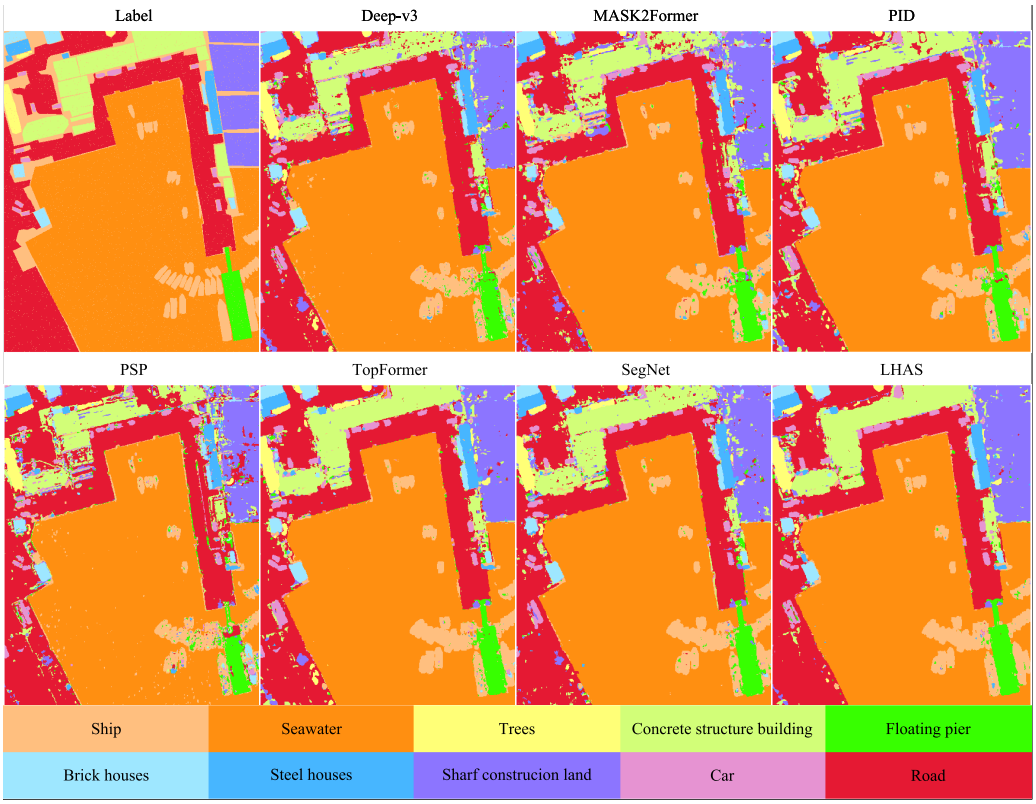


Fig. 7. PingAn's ground-truth and segmentation maps in different methods.

TABLE XIII
INFERENCE TIME OF DIFFERENT METHODS ON HANCHUAN AND LONGKOU DATASETS

Method	HanChuan (s)	LongKou (s)	PingAn (s)
Deep-v3	8	6	20
MASK2former	8	6	20
PID	7	5	18
PSP	8	6	20
TopFormer	7	5	18
SegNet	8	6	19
LHAS-P	8	6	19
LHAS	7	5	17

in Tables X–XII. This is because the multiscale information of Deep-v3 is not fully utilized. In contrast, the segmentation performance of PID is acceptable, but the segmentation effect is not smooth enough. However, the resulting graphs generated by LHAS are more accurate, spatially smoother, and have fewer mislabeled pixels than the other methods, which is consistent with the results reported in Tables X–XII.

F. Computational Analysis on Embedded Device

In the actual application scenario, the derivation model can remove the PAM, which reduces the computational complexity and operation time. To illustrate the computational complexity of the proposed LHAS, using PyTorch in Python on an NVIDIA Jetson AGX Orin embedded device, Table XIII provides the inference time of various methods on the above datasets. For all three datasets, LHAS is the fastest. This inference speed is faster than the hyperspectral imaging process, so it can meet real-time application scenarios. Although all the compared methods can achieve inference speed on the order of seconds, LHAS is the fastest method with the best classification performance. On the whole, the inference time and inference effect are acceptable.

IV. CONCLUSION

In this article, we propose GDA for GDA to meet the data requirements of HSI semantic segmentation. At the same time, a lightweight para-block is used for feature extraction, hierarchical attention is used for multiscale information fusion, and the PAM is used for clustering and contrastive learning to enhance the differences between different categories and further improve network performance. To reduce the complexity of LHAS, the PAM can be removed directly during derivation. Experimental results on HSI datasets in three different scenarios show that LHAS outperforms the existing state-of-the-art HSI semantic segmentation methods, which verifies the effectiveness of LHAS.

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